

# Understanding Syntax: Python and R Compared

## A Comparative Introduction for Data Analysis

true

### Abstract

This handout complements the presentation *Understanding Syntax: Python and R Compared*. It explains the most important syntax differences between R and Python for introductory data analysis. The focus is on variables, data structures, indexing, filtering, data manipulation, and translation between R's `dplyr` style and Python's `pandas` style. The aim is not to decide which language is better, but to help students understand how the same analytical logic can be expressed differently in each language.

### GitHub Repository:

<https://github.com/Sri-prog-7943/Python-and-R-Compared>

## 1. Introduction

R and Python are two of the most widely used programming languages in data analysis. R was developed with a strong focus on statistical computing, data analysis, and visualisation. Python, by contrast, is a general-purpose programming language that has become highly important in data science through libraries such as `pandas`, `NumPy`, and `scikit-learn` (McKinney, 2022; Python Software Foundation, 2024).

For students in data analysis, the most important point is that both languages often solve the same analytical problem. The difference is mainly the syntax: how the instruction is written. Therefore, this handout explains the similarities and differences between the two languages using short practical examples.

The central learning objective is:

Students should be able to read simple R and Python code, identify equivalent syntax patterns, and translate basic data manipulation operations between the two languages.

## 2. Python and R at a Glance

The following table summarises the most important differences.

| Feature              | R                                             | Python                                       |
|----------------------|-----------------------------------------------|----------------------------------------------|
| Main purpose         | Statistical computing and graphics            | General-purpose programming and data science |
| Main data structure  | <code>data.frame</code> / <code>tibble</code> | <code>pandas.DataFrame</code>                |
| Assignment           | <code>x &lt;- 10</code>                       | <code>x = 10</code>                          |
| First index          | 1                                             | 0                                            |
| Popular data package | <code>dplyr</code> / <code>tidyverse</code>   | <code>pandas</code>                          |
| Typical workflow     | Pipe-based analysis                           | Method chaining / object-based analysis      |

Although the syntax looks different, the analytical thinking is often the same. For example, both languages can import data, clean data, filter rows, calculate summary statistics, and create visualisations. The user only needs to understand how each language expresses these tasks.

### 3. Assignment and Variables

A variable stores information that can be reused later. In R, the preferred assignment operator is `<-`, although `=` also works. In Python, assignment is done with `=`.

#### R

```
student_name <- "Srikumar"
score <- 92.5
is_passed <- TRUE

student_name
```

```
## [1] "Srikumar"
```

#### Python

```
student_name = "Srikumar"
score = 92.5
is_passed = True

print(student_name)
```

A small but important difference is that R uses uppercase logical values, such as `TRUE` and `FALSE`. Python uses `True` and `False`. Python is also stricter about indentation, especially in control structures such as `if` statements and loops.

### 4. Data Structures

Data structures organise information. In introductory data analysis, the most important structure is the data frame. A data frame is a table with rows and columns, similar to an Excel sheet.

#### R data frame

```
##   name score  program
## 1 Alice   88   Finance
## 2  Bob   76 Marketing
## 3 Carol  95   Finance
```

#### Python DataFrame

```
import pandas as pd

# Create DataFrame
students = pd.DataFrame({
    "name": ["Alice", "Bob", "Carol"],
    "score": [88, 76, 95],
    "program": ["Finance", "Marketing", "Finance"]
})

# Display the DataFrame
print(students)
```

In R, a data frame is available in base R. In Python, the standard data frame structure usually comes from the `pandas` library (The pandas Development Team, 2024). This means that Python users normally import `pandas` before creating or analysing tabular data.

## 5. Indexing: One-Based vs. Zero-Based Counting

One of the most common mistakes when moving between R and Python is indexing. R starts counting at 1, while Python starts counting at 0.

**R**

```
## [1] 85
```

```
## [1] 90
```

**Python**

```
## 85
```

```
## 90
```

This distinction is small, but it can create errors in practical analysis. When translating code, the index position must always be checked carefully.

## 6. Filtering and Selecting Data

Filtering means keeping only rows that satisfy a condition. Selecting means keeping only specific columns.

### 6.1 Filtering Rows

**R with dplyr**

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

##   name score program
## 1 Alice    88 Finance
## 2 Carol    95 Finance
```

**Python with pandas**

```
import pandas as pd

students = pd.DataFrame({
    "name": ["Alice", "Bob", "Carol"],
    "score": [88, 76, 95],
    "program": ["Finance", "Marketing", "Finance"]
})

students[students["score"] > 80]
```

In R, `filter()` is commonly used in the `dplyr` package. In Python, row filtering often uses Boolean indexing, where the condition is placed inside square brackets.

### 6.2 Selecting Columns

**R**

```
students |>
  select(name, score)
```

## Python

```
students[["name", "score"]]
```

The double square brackets in Python are important because a list of column names is passed into the DataFrame.

## 7. Data Manipulation with dplyr and pandas

R's dplyr package is popular because it makes code readable. It uses verbs such as `filter()`, `select()`, `mutate()`, `summarise()`, and `arrange()` (Wickham & Grolemund, 2017). Python's pandas library performs similar tasks, but the syntax is object-oriented.

| Task              | R / dplyr                | Python / pandas                                        |
|-------------------|--------------------------|--------------------------------------------------------|
| Add a column      | <code>mutate()</code>    | <code>.assign()</code> or <code>df["new"] = ...</code> |
| Filter rows       | <code>filter()</code>    | Boolean indexing or <code>.query()</code>              |
| Sort rows         | <code>arrange()</code>   | <code>.sort_values()</code>                            |
| Group data        | <code>group_by()</code>  | <code>.groupby()</code>                                |
| Calculate summary | <code>summarise()</code> | <code>.agg()</code> or <code>.mean()</code>            |

## 8. Complete Translation Example

The following example shows how the same task can be written in R and Python. The task is to calculate price per unit, keep only products with revenue above 10,000, and sort by revenue.

### R Code

```
##   product revenue units price_per_unit
## 1      C   22000   200      110.0000
## 2      A   15000   120      125.0000
## 3      D   11000    95      115.7895
```

### Python Code

```
import pandas as pd

sales = pd.DataFrame({
    "product": ["A", "B", "C", "D"],
    "revenue": [15000, 8500, 22000, 11000],
    "units": [120, 85, 200, 95]
})

result = (
    sales
    .assign(price_per_unit = sales["revenue"] / sales["units"])
    .query("revenue > 10000")
    .sort_values("revenue", ascending=False)
    .reset_index(drop=True)
)

print(result)
```

The example demonstrates that the analytical logic is identical. First, a new column is created. Second, rows are filtered. Third, the output is sorted. The main difference is how each language expresses these steps.

### **Reflection questions**

1. Which R function is equivalent to `.assign()` in Python?
2. Why does Python require `df["column"]` instead of only writing the column name?
3. What mistake could happen because R starts at 1 and Python starts at 0?

## Conclusion

The comparison between R and Python shows that syntax is a surface-level difference, while the underlying data analysis logic is often the same. A student who understands how to filter, select, group, and summarise data in one language can transfer that knowledge to the other language.

The most important points are:

- R is especially strong for statistics, academic data analysis, and visualisation.
- Python is highly flexible and widely used for data science, automation, and machine learning.
- R starts indexing at 1; Python starts indexing at 0.
- R's pipe-based syntax is similar in purpose to pandas method chaining.
- Learning both languages improves analytical flexibility and makes collaboration easier.

For this course, R remains the primary language because the module focuses on data analysis and decision-making using R. However, understanding Python is useful because many business analytics environments use both tools.

## Affidavit

I hereby affirm that this submitted paper was authored unaided and solely by me. Additionally, no other sources than those in the reference list were used. Parts of this paper, including tables and figures, that have been taken either verbatim or analogously from other works have in each case been properly cited with regard to their origin and authorship. This paper either in parts or in its entirety, be it in the same or similar form, has not been submitted to any other examination board and has not been published.

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*28 April 2026,*

*Rösrath*

McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and jupyter* (3rd ed.). O'Reilly Media.

Python Software Foundation. (2024). *The python language reference*. <https://docs.python.org/3/reference/>

The pandas Development Team. (2024). *Pandas documentation*. <https://pandas.pydata.org/docs/>

Wickham, H., & Grolemund, G. (2017). *R for data science: Import, tidy, transform, visualize, and model data*. O'Reilly Media. <https://r4ds.had.co.nz/>